

Lecture 19

BACKTRACKING SEARCH

FOR CSPS

Artificial Intelligence
COSC-3112

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Agenda

- Real World CSPs
- Standard Search Formulation
- Improving Backtracking Efficiency
- Most Constrained Variable
- Least Constrained Variable
- Forward Checking

CSP

- Represent a state with a set of variable/value pairs
- Represent the conditions for a solution by a set of constraints on the variables
- CSP consists of 3 components:
 - X is a set of variables, $\{X_1, \dots, X_n\}$.
 - D is a set of domains, $\{D_1, \dots, D_n\}$, one for each variable.
 - C is a set of constraints that specify allowable combination of values.

CSP Assignment

- Each state in a CSP is defined by an assignment of values to some of the variables, $\{X_i=v_i, X_j=v_j, \dots\}$

OR

- A *state* is an *assignment* of values to some or all variables.
- Types of Assignments
 - Legal Assignment
 - Partial Assignment
 - Complete Assignments

CSP Assignment

- **Consistent Assignment**

- An assignment that does not violate any constraints is called a **consistent** or legal assignment
 - Example
 - Map Coloring CSP

- **Complete Assignment**

- An assignment is ***complete*** when every variable has a value.

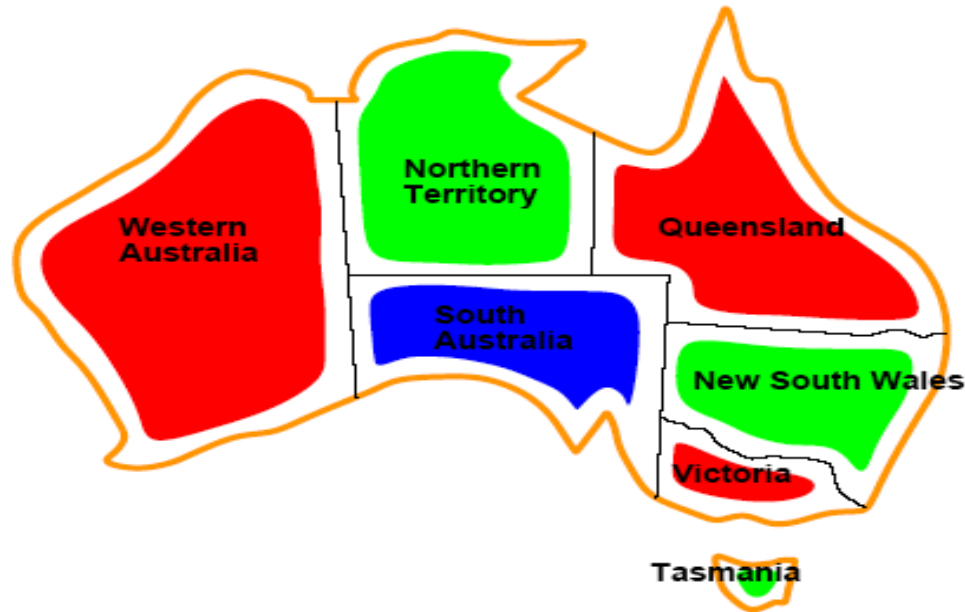
- **Partial Assignment**

- An assignment is ***partial*** when some variables have no values.

Solution to CSP

- A ***solution*** to a CSP is a complete and consistent assignment.
- Some CSPs require a solution that maximizes an *objective function*.

CSP example: map coloring

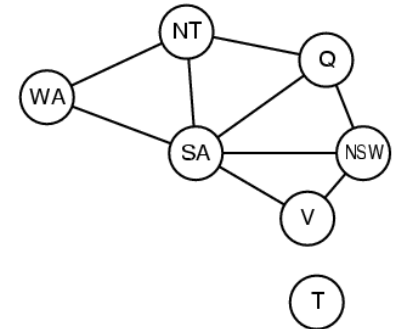


- Solutions are assignments satisfying all constraints, e.g.
 $\{WA=red, NT=green, Q=red, NSW=green, V=red, SA=blue, T=green\}$

Real-world CSPs

- Many CSPs cannot be solved by **inference** alone
 - There comes a time when we must **search** for a solution
- Examples
 - Assignment problems
 - e.g., who teaches what class
 - Timetabling problems
 - e.g., which class is offered when and where?
 - Transportation scheduling
 - Factory scheduling
- Notice that many real-world problems involve real-valued variables

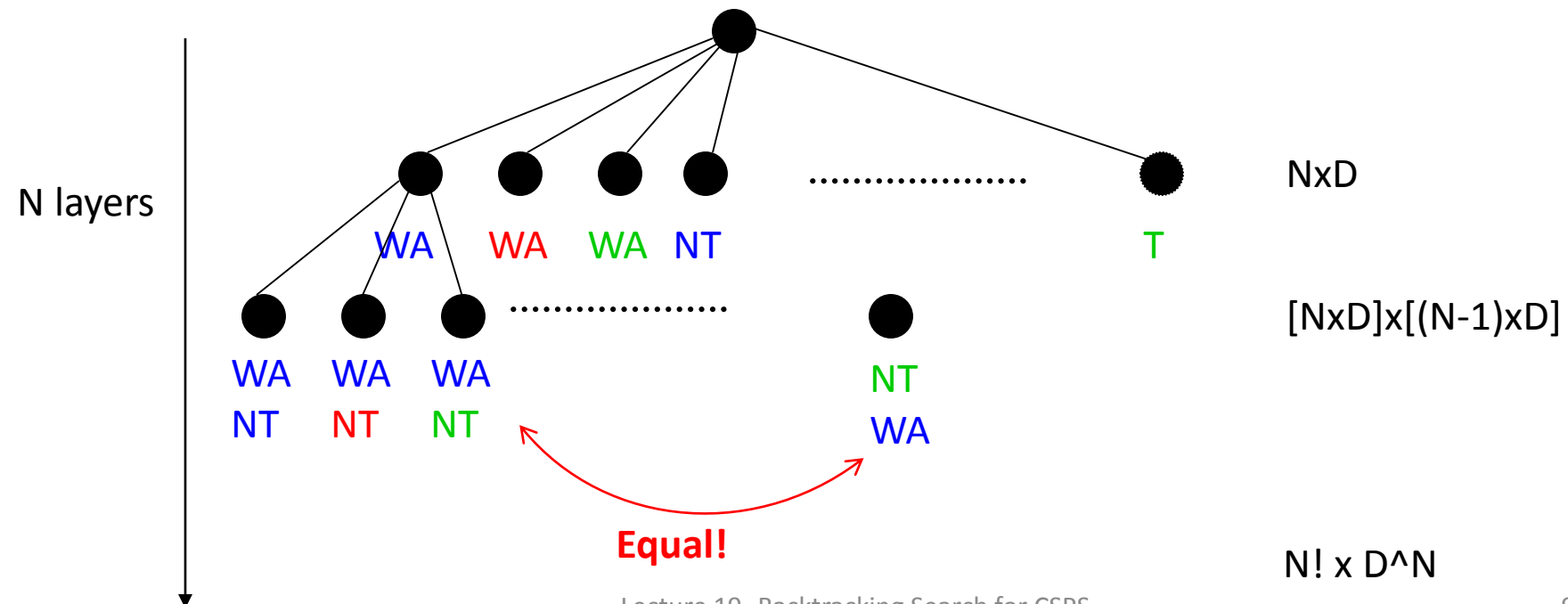
Standard Search Formulation



Let's try the standard search formulation.

We need:

- Initial state: none of the variables has a value (color)
- Successor state: one of the variables without a value will get some value.
- Goal: all variables have a value and none of the constraints is violated.



Lecture 19- Backtracking Search for CSPs
There are $N! \times D^N$ nodes in the tree but only D^N distinct states

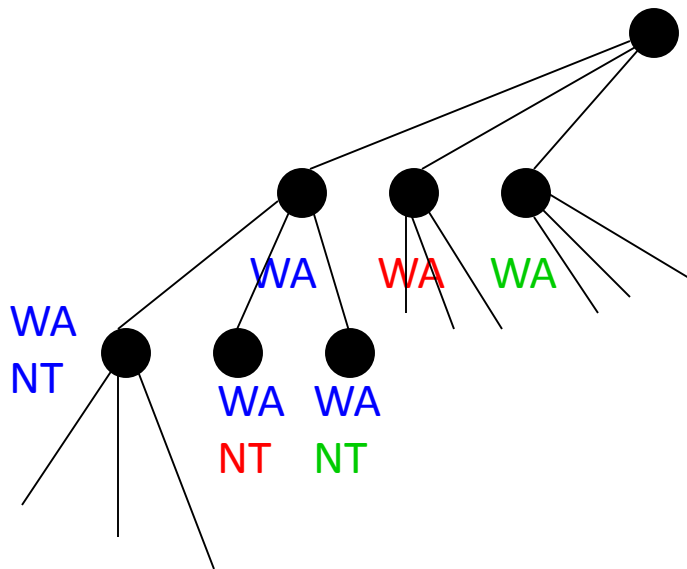
CSP as a Standard Search Problem

- A CSP can easily be expressed as a standard search problem.
- Incremental formulation
 - *Initial State*: the empty assignment $\{\}$
 - *Successor function*: Assign a value to an unassigned variable provided that it does not violate a constraint
 - *Goal test*: the current assignment is complete
(by construction it is consistent)
 - *Path cost*: constant cost for every step (not really relevant)

Backtracking (Depth-First) search

- Special property of CSPs: They **are commutative**:
This means: the order in which we assign variables does not matter.
- Better search tree: First **order** variables, then assign them values **one-by-one**.

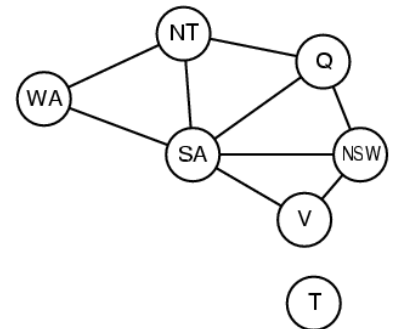
$$\begin{matrix} \text{NT} \\ \text{WA} \end{matrix} = \begin{matrix} \text{WA} \\ \text{NT} \end{matrix}$$



D

D^2

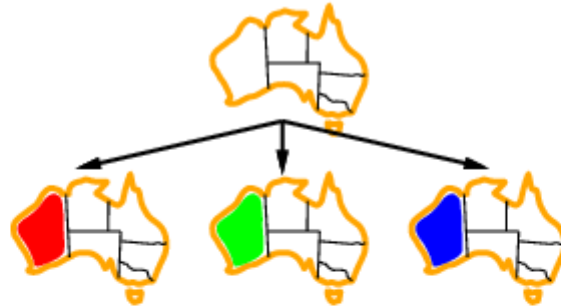
D^N



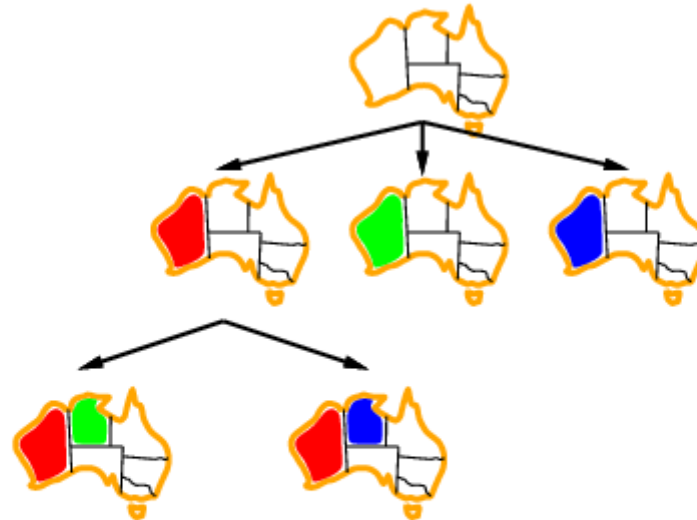
Backtracking example



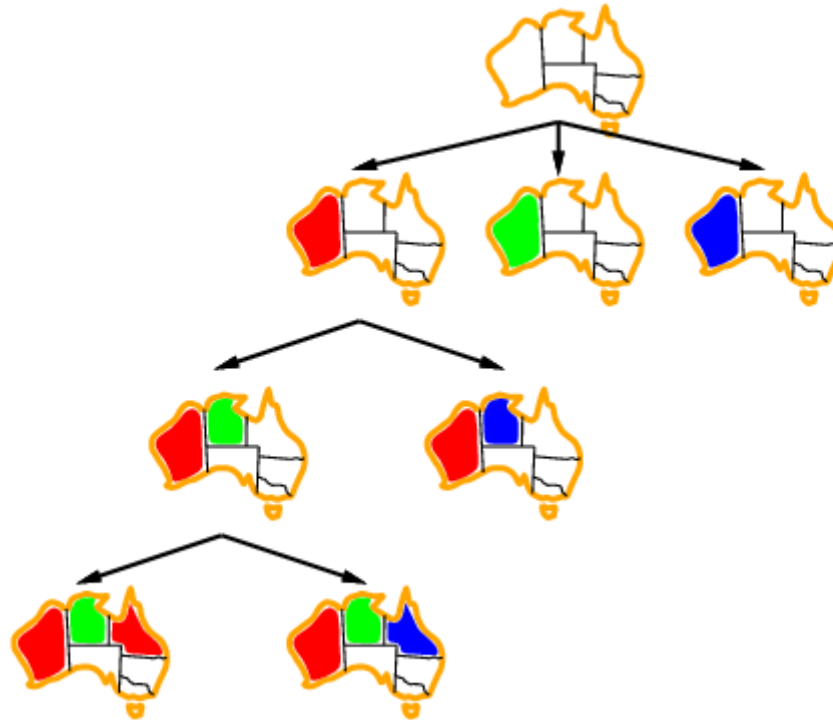
Backtracking example



Backtracking example



Backtracking example

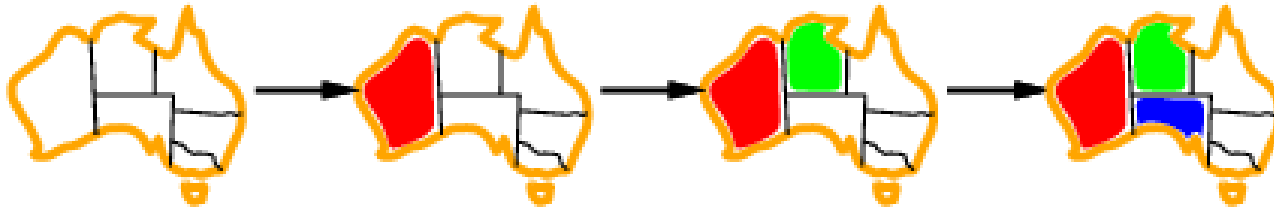


Improving Backtracking Efficiency

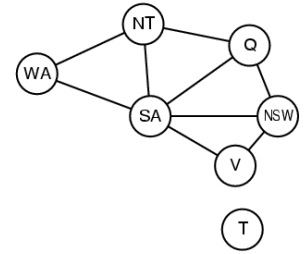
- **General-purpose** methods can give huge gains in speed:
 - Which variable should be assigned next?
 - In what order should its values be tried?
 - What inferences should be performed at each step in the search?
 - Can we detect inevitable failure early?

Most constrained variable

- Most constrained variable:
 - choose the variable with the fewest legal values

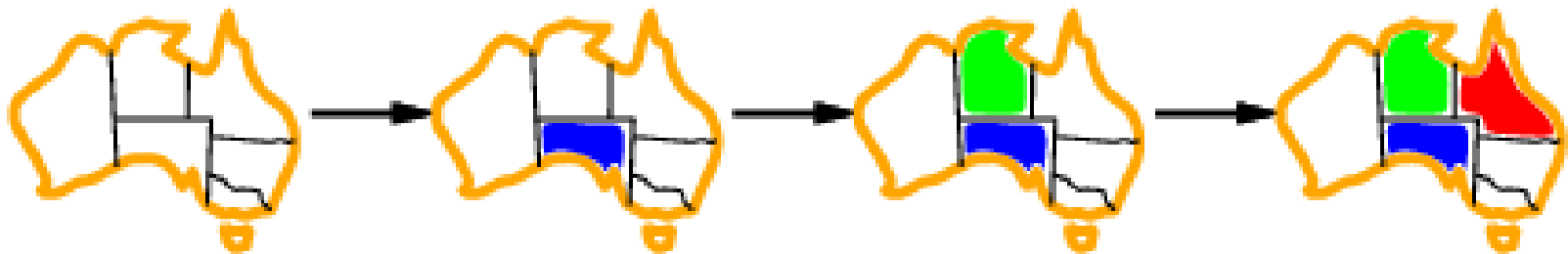


- a.k.a. **minimum remaining values (MRV)** heuristic
- Picks a variable which will cause failure as soon as possible, allowing the tree to be pruned.
 - Most likely to fail soon (so prunes pointless searches)



Most constraining variable

- Tie-breaker among most constrained variables
 - **Tie**:- Such as choosing start state
- Most constraining variable:
 - choose the variable involved in the most constraints (**degree heuristic**)
 - having most branches
 - E.g., in the map example, SA adjacent to the most states.
 - **Reduces branching factor, since fewer legal successors of that node**



Least constraining value



- *Given a variable*, choose the least constraining value:
 - the one that rules out the fewest values in the remaining variables

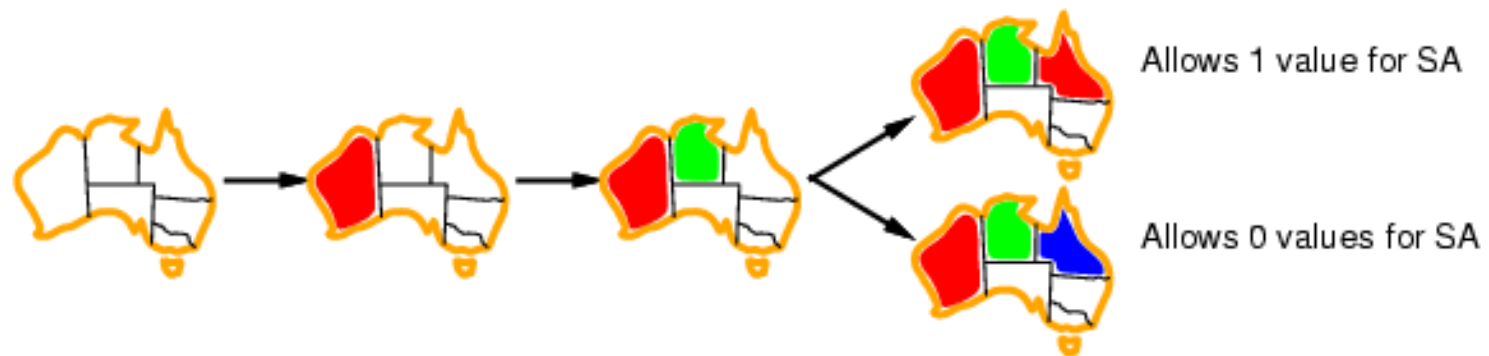


- Leaves maximal flexibility for a solution.
 - You've chosen the variable, now let's make the most of it
- Combining these heuristics makes 1000 queens feasible

Least constraining value

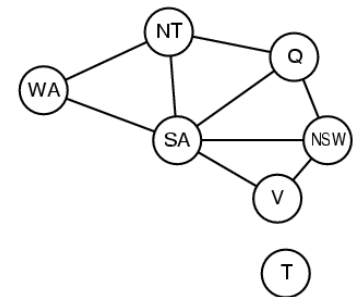
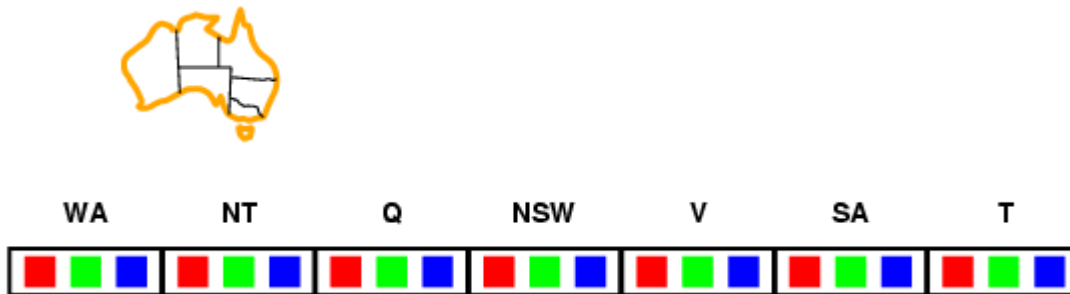
$var \leftarrow \text{SELECT-UNASSIGNED-VARIABLE}(assignment, csp)$

- Before the assignment to the rightmost state: one region has one remaining; one region has two; three regions have three. Choose the region with only one remaining



Forward checking

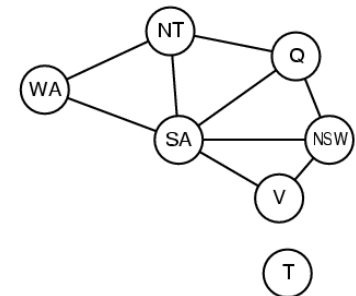
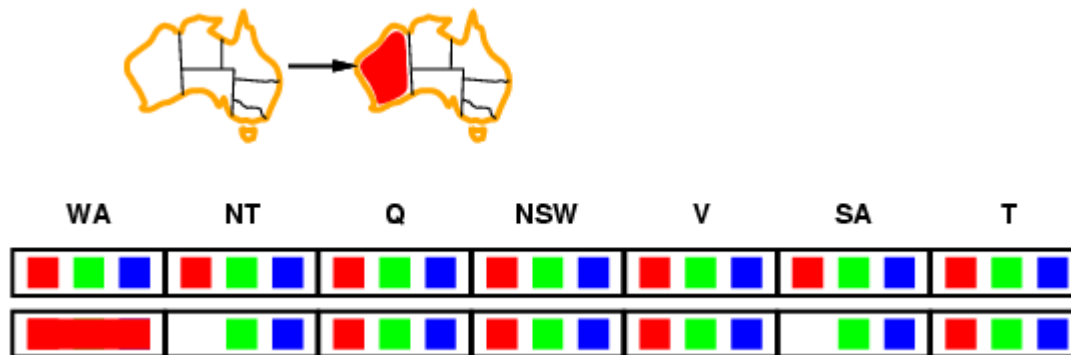
- Idea:
 - Keep track of remaining legal values for unassigned variables
 - Terminate search when any variable has no legal values



Forward checking

- Idea:

- Keep track of remaining legal values for unassigned variables
- Terminate search when any variable has no legal values



Assign {WA=red}

Effects on other variables connected by constraints to WA

NT can no longer be red

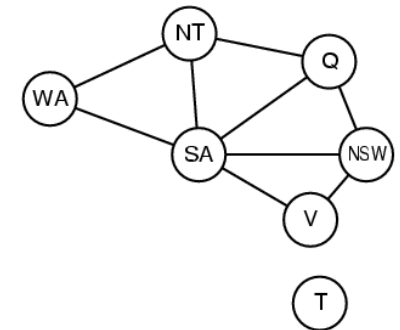
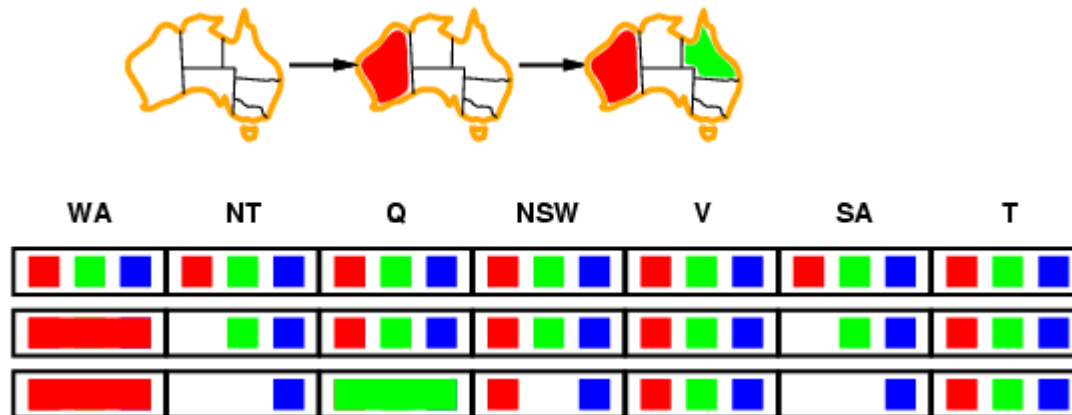
SA can no longer be red

Note: this example is not using MRV; if it were, we would choose NT or SA next. But, we will choose Q next. This example is from the text. It shows the example here, then talks through what would happen if we had used MRV

Forward checking

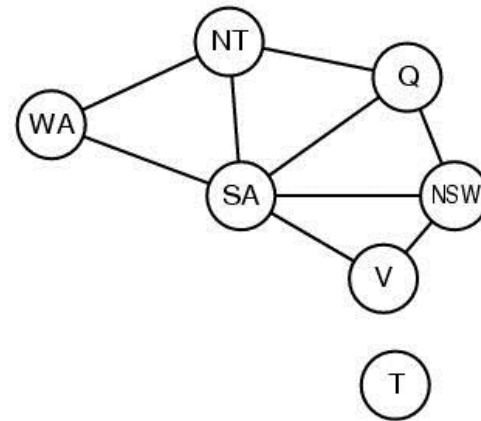
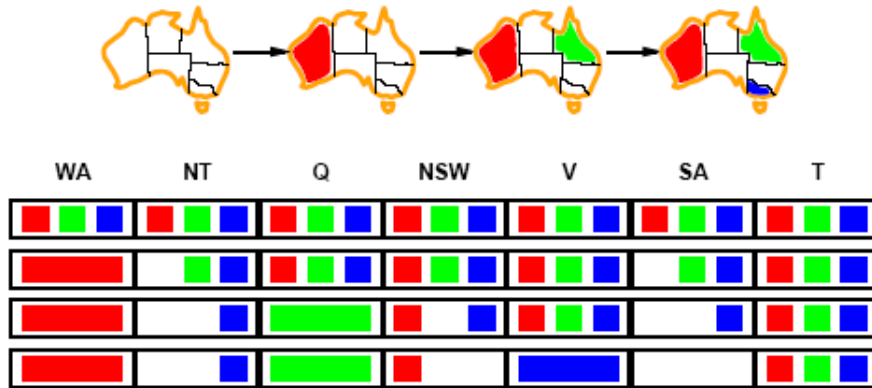
- Idea:

- Keep track of remaining legal values for unassigned variables
- Terminate search when any variable has no legal values



- Assign $\{Q=green\}$
- Effects on other variables connected by constraints with WA
 - NT can no longer be green*
 - NSW can no longer be green*
 - SA can no longer be green*

Forward checking



- Assign $\{V=blue\}$
- Effects on other variables connected by constraints with WA
 - *NSW can no longer be blue*
 - *SA is empty*
- Forward Checking has detected that partial assignment is *inconsistent* with the constraints and backtracking can occur.